

ft_linux — Package Analysis Against LFS Book 12.4

Cross-reference of the mandatory package list from the ft_linux subject (v3.6) against LFS Book 12.4. Packages are split into three sections:

1. **Installed** – present in LFS 12.4, with verification hints.
2. **Deprecated / Removed / Replaced** – present in the subject but no longer part of LFS 12.4, with an explanation.
3. **Renamed / Rebranded** – same upstream software, different name.

Note

LFS 12.4 ships in two editions: *SysV* and *Systemd*. Notes that differ between the two are marked accordingly. The subject package list (Syslogd + Sysvinit) targets a SysV-style system. I went for a SysV build.

Installed Packages

These packages are present in LFS 12.4 exactly as listed in the subject. Because LFS has no package manager by default, the canonical way to confirm installation is to query the installed binary or library directly. Some executables do not have a version flag and must be run to expose the usage tooltip, others are libraries and don't expose have executables at all. For those, please list the files to prove they are installed.

Acl

Verify: `getfacl --version`
Path: `/usr/bin/getfacl, /usr/lib/libacl.so`

Attr

Verify: `attr`
Path: `/usr/bin/attr, /usr/lib/libattr.so`

Autoconf

Verify: `autoconf --version`
Path: `/usr/bin/autoconf`

Automake

Verify: `automake --version`
Path: `/usr/bin/automake`

Bash

Verify: `bash --version`

Path: /bin/bash

Bc

Verify: bc --version

Path: /usr/bin/bc

Binutils

Verify: ld --version, as --version

Path: /usr/bin/ld, /usr/bin/as, /usr/lib/libbfd*.so

Bison

Verify: bison --version

Path: /usr/bin/bison

Bzip2

Verify: bzip2 --version

Path: /usr/bin/bzip2, /usr/lib/libbz2.so

Check

Verify: pkg-config --modversion check (or ls /usr/lib/libcheck*)

Path: /usr/lib/libcheck.so

Note: Not present in LFS 12.4 anymore. Installed for the sake of it the the correction.

Coreutils

Verify: ls --version (ls is provided by coreutils)

Path: /usr/bin/ls, /usr/bin/cp, /usr/bin/chmod, etc.

DejaGNU

Verify: runtest --version

Path: /usr/bin/runtest

Note: Used only for running the GCC and Binutils test suites; not needed after the build.

Diffutils

Verify: diff --version

Path: /usr/bin/diff

E2fsprogs

Verify: /usr/sbin/e2fsck

Path: /usr/sbin/e2fsck, /usr/lib/libe2p.so, /usr/lib/libext2fs.so

Expat

Verify: xmlwf --version (ships with expat) or ls /usr/lib/libexpat*
Path: /usr/lib/libexpat.so

Expect

Verify: expect -v
Path: /usr/bin/expect, /usr/lib/expect*/
Note: Used only during test suites (GCC, Binutils); can be absent after the build without affecting the running system.

File

Verify: file --version
Path: /usr/bin/file, /usr/lib/libmagic.so

Findutils

Verify: find --version
Path: /usr/bin/find, /usr/bin/locate

Flex

Verify: flex --version
Path: /usr/bin/flex

Gawk

Verify: gawk --version
Path: /usr/bin/gawk

GCC

Verify: gcc --version, g++ --version
Path: /usr/bin/gcc, /usr/lib/libgcc_s.so, /usr/lib/libstdc++.so

GDBM

Verify: ls /usr/lib/libgdbm*
Path: /usr/lib/libgdbm.so, /usr/include/gdbm.h

Gettext

Verify: gettext --version
Path: /usr/bin/gettext, /usr/lib/libintl.so

Glibc

Verify: `/usr/lib/libc.so.6` (execute it directly to see version)
Path: `/usr/lib/libc.so.6, /usr/lib/ld-linux-x86-64.so.2`

GMP

Verify: `ls /usr/lib/libgmp*`
Path: `/usr/lib/libgmp.so, /usr/include/gmp.h`

Gperf

Verify: `gperf --version`
Path: `/usr/bin/gperf`

Grep

Verify: `grep --version`
Path: `/usr/bin/grep`

Groff

Verify: `groff --version`
Path: `/usr/bin/groff`

GRUB

Verify: `/usr/sbin/grub-install --version`
Path: `/usr/sbin/grub-install, /boot/grub/`

Gzip

Verify: `gzip --version`
Path: `/usr/bin/gzip`

Iana-Etc

Verify: `cat /etc/services | head -n4, cat /etc/protocols | head -n4`
Path: `/etc/services, /etc/protocols`
Note: This package is purely data files; there are no binaries.

Inetutils

Verify: `hostname --version`
Path: `/usr/bin/hostname, /usr/bin/ping, /usr/bin/ftp`

IPRoute2

Verify: /usr/sbin/ss --version
Path: /usr/sbin/ip, /usr/sbin/ss

Kbd

Verify: loadkeys --version
Path: /usr/bin/loadkeys, /usr/bin/setfont

Kmod

Verify: kmod --version
Path: /usr/bin/kmod; also provides lsmod, modprobe, insmod as symlinks.

Less

Verify: less --version
Path: /usr/bin/less

Libcap

Verify: /usr/sbin/capsh --version (or ls /usr/lib/libcap*)
Path: /usr/lib/libcap.so, /usr/sbin/setcap

Libpipeline

Verify: ls /usr/lib/libpipeline*
Path: /usr/lib/libpipeline.so
Note: Used by man-db; no standalone binary.

Libtool

Verify: libtool --version
Path: /usr/bin/libtool

M4

Verify: m4 --version
Path: /usr/bin/m4

Make

Verify: make --version
Path: /usr/bin/make

Man-DB

Verify: man --version
Path: /usr/bin/man, /usr/bin/mandb

Man-pages

Verify: `man 2 open` (check a syscall man page)
Path: `/usr/share/man/man2/, /usr/share/man/man3/`
Note: Data files only; no binaries.

MPC

Verify: `ls /usr/lib/libmpc*`
Path: `/usr/lib/libmpc.so, /usr/include/mpc.h`

MPFR

Verify: `ls /usr/lib/libmpfr*`
Path: `/usr/lib/libmpfr.so, /usr/include/mpfr.h`

Ncurses

Verify: `ls /usr/lib/libncurses*`
Path: `/usr/lib/libncurses.so, /usr/bin/tput`

Patch

Verify: `patch --version`
Path: `/usr/bin/patch`

Perl

Verify: `perl --version`
Path: `/usr/bin/perl, /usr/lib/perl5/`

Procps (procps-ng)

Verify: `ps --version`
Path: `/usr/bin/ps, /usr/bin/top, /usr/lib/libprocps.so`
Note: The subject calls it "Procps"; in LFS 12.4 it is explicitly *procps-ng*, the maintained fork. Functionally identical for the purposes of this project.

Psmisc

Verify: `pstree --version`
Path: `/usr/bin/pstree, /usr/bin/fuser, /usr/bin/killall`

Readline

Verify: `ls /usr/lib/libreadline*`
Path: `/usr/lib/libreadline.so, /usr/include/readline/`

Sed

Verify: `sed --version`
Path: `/usr/bin/sed`

Shadow

Verify: `/usr/sbin/useradd`
Path: `/usr/sbin/useradd, /usr/bin/passwd`

Sysklogd

Verify: `/sbin/syslogd -v` (or `ls /sbin/syslogd`)
Path: `/sbin/syslogd, /sbin/klogd`
Note: **SysV edition only.** If you built the Systemd edition of LFS 12.4, `journald` from `systemd` replaces `syslogd` entirely and this package is absent.

Sysvinit

Verify: `/sbin/init --version`
Path: `/sbin/init`
Note: **SysV edition only.** Not present in the Systemd edition; `systemd` provides `/sbin/init` as a symlink to itself instead.

Tar

Verify: `tar --version`
Path: `/usr/bin/tar`

Tcl

Verify: `tclsh` (type `info patchlevel` at the prompt)
Path: `/usr/bin/tclsh*, /usr/lib/libtcl*.so`
Note: Required only to run DejaGNU / Expect test suites; not needed on the final system but is included in LFS 12.4 nonetheless.

Texinfo

Verify: `makeinfo --version` (or `texi2any --version`)
Path: `/usr/bin/makeinfo`

Time Zone Data (tzdata)

Verify: `ls /usr/share/zoneinfo/`
Path: `/usr/share/zoneinfo/, /etc/localtime`
Note: Data files only; installed by the Glibc chapter in LFS 12.4.

Util-linux

Verify: mount --version

Path: /usr/bin/mount, /usr/bin/fdisk, /usr/bin/lsblk,
/usr/lib/libuuid.so, /usr/lib/libblkid.so

Vim

Verify: vim --version

Path: /usr/bin/vim

XML::Parser

Verify: head /usr/lib/perl5/5.42/site_perl/XML/Parser.pm | head -n7"

Path: head /usr/lib/perl5/5.42/site_perl/XML/Parser.pm

Note: Perl module.

Xz Utils

Verify: xz --version

Path: /usr/bin/xz, /usr/lib/liblzma.so

Zlib

Verify: ls /usr/lib/libz*

Path: /usr/lib/libz.so, /usr/include/zlib.h

Deprecated / Removed / Replaced Packages

These packages appear in the subject's mandatory list but are **not** installed by LFS 12.4, either because they have been removed from the book, superseded by another package, or absorbed into a different source tarball.

Eudev

Status: **Replaced** by udev from the systemd source tarball.

Subject ref: Listed as the kernel-module loader (equivalent to udev).

What LFS 12.4 does: Starting with LFS 11.x, the book dropped the standalone *Eudev* fork. In the **SysV edition**, LFS 12.4 downloads the *systemd* source tarball and builds *only* the udev component from it (/usr/lib/udev/, /usr/bin/udevadm). In the **Systemd edition**, full systemd is used and udev is part of it. Eudev, which was a community fork of udev intended for non-systemd systems, is no longer necessary because the build procedure extracts the udev piece directly.

Verify (LFS 12.4 SysV): udevadm --version and ls /usr/lib/udev/rules.d/

Intltool

Status: **Removed** from LFS (dropped around LFS 9.x).

Subject ref: Listed as a required package in the mandatory part.

Why removed: Intltool was a Perl-based toolchain for extracting translatable strings from source files and merging them with gettext .po files. Modern projects have migrated to using gettext autotools macros or the xgettext/msgfmt workflow directly, making Intltool redundant. The LFS editors removed it once no remaining LFS package required it as a build dependency. You may encounter it in BLFS (some older GTK applications still need it), but it is absent from the base LFS 12.4 build.

Pkg-config

Status: Replaced by *pkgconf* since LFS 12.0.
Subject ref: Listed under its original name "Pkg-config".
Why replaced: The original pkg-config binary (freedesktop.org) has seen minimal maintenance for years. LFS 12.0 switched to *pkgconf*, an independently written, fully compatible, and actively maintained replacement. pkgconf installs a /usr/bin/pkg-config symlink, so all existing scripts and build systems remain unaffected.
Verify: pkg-config --version (reports pkgconf's version number)
Path: /usr/bin/pkgconf, symlinked as /usr/bin/pkg-config

Udev-lfs Tarball

Status: Deprecated / Removed.
Subject ref: Listed as a distinct package "Udev-lfs Tarball".
Why removed: This was a small, LFS-project-maintained tarball containing supplementary udev rules and helper shell scripts that were not shipped with either eudev or upstream udev. It filled the gap when LFS was transitioning between udev versions. As of LFS 11.x, all required udev rules are either upstream (in the systemd/eudev source itself) or generated at runtime by udevadm. The separate tarball became unnecessary and was dropped from the book entirely.

Summary Table

Package	Status	Notes
Acl	Installed	
Attr	Installed	
Autoconf	Installed	
Automake	Installed	
Bash	Installed	
Bc	Installed	
Binutils	Installed	
Bison	Installed	
Bzip2	Installed	
Check	Installed	
Coreutils	Installed	
DejaGNU	Installed	Test-suite tool only
Diffutils	Installed	

Eudev	REPLACED	udev from systemd tarball used instead
E2fsprogs	Installed	
Expat	Installed	
Expect	Installed	Test-suite tool only
File	Installed	
Findutils	Installed	
Flex	Installed	
Gawk	Installed	
GCC	Installed	
GDBM	Installed	
Gettext	Installed	
Glibc	Installed	
GMP	Installed	
Gperf	Installed	
Grep	Installed	
Groff	Installed	
GRUB	Installed	
Gzip	Installed	
Iana-Etc	Installed	Data files only (/etc/services, /etc/ protocols)
Inetutils	Installed	
Intltool	REMOVED	Dropped ~LFS 9.x; gettext covers its role
IPRoute2	Installed	
Kbd	Installed	
Kmod	Installed	
Less	Installed	
Libcap	Installed	
Libpipeline	Installed	
Libtool	Installed	
M4	Installed	
Make	Installed	
Man-DB	Installed	
Man-pages	Installed	Data files only
MPC	Installed	
MPFR	Installed	
Ncurses	Installed	
Patch	Installed	
Perl	Installed	
Pkg-config	REPLACED	pkgconf used; provides pkg-config symlink
Procps	Installed	Ships as procps-ng in LFS 12.4

Psmisc	Installed	
Readline	Installed	
Sed	Installed	
Shadow	Installed	
Sysklogd	Installed	SysV edition only
Sysvinit	Installed	SysV edition only
Tar	Installed	
Tcl	Installed	Test-suite tool only
Texinfo	Installed	
Time Zone Data	Installed	Installed as part of Glibc chapter
Udev-lfs Tarball	REMOVED	Rules now upstream in systemd/eudev
Util-linux	Installed	
Vim	Installed	
XML::Parser	Installed	Perl module
Xz Utils	Installed	
Zlib	Installed	